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Genetic Algorithm

Theory of evolution

The theory of evolution is the basic idea behind Genetic Algorithm.

According to Darwin's theory,
Fit individuals get together and make fitter individuals.

→ Two good solutions are combined to make better solutions.

Evolutionary optimization.

1. Evolutionary computing
2. Combinatorial optimization
3. Genetic algorithms.

Evolutionary computing

Evolutionary computing is a problem solving approach inspired by natural evolution.

It uses concepts such as :

- Selection
- Reproduction
- Crossover
- Mutation
- Fitness.

* The aim is to gradually produce better solutions.

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sector AI

Combinational optimization

The process of finding the best solution from a large number of possible combinations.

Ex:
- Shortest route problem
- Scheduling
- Travelling salesman problem
- Resource allocation
- Job assignment.

Genetic Algorithms

A genetic algorithm is an optimization method that uses selection, crossover and mutation to find better solutions.

Key terms

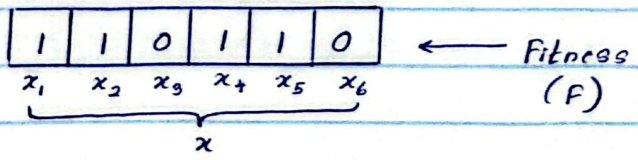
- Population : A group of possible solutions.
- Chromosome : Representation of one solution.
- Gene : A part of a chromosome
- Fitness : How good a solution is.
- Selection : Choosing good solutions.
- Crossorer : Combining two solutions.
- Mutation : Randomly changing a solution.
- Generation : One complete cycle of the algorithm.

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Genetic Algorithm

1. Decide the population size
2. Decide the chromosome structure
3. Decide the fitness
4. 4.1 Selection : Choose good solutions from the current population.
   - Roulette wheel selection
   - Binary tournament
   - 4.2 Crossover : Combine two selected parent solutions to create children
   - 4.3 Mutation : Randomly change some genes in the children
   - More children to make ? — Yes → back to 4.1 Selection
   - No
5. Make the new population
   - Stopping Criteria met ? — No → back to 4.1 Selection
   - Yes
   - Stop

Chromosome representation

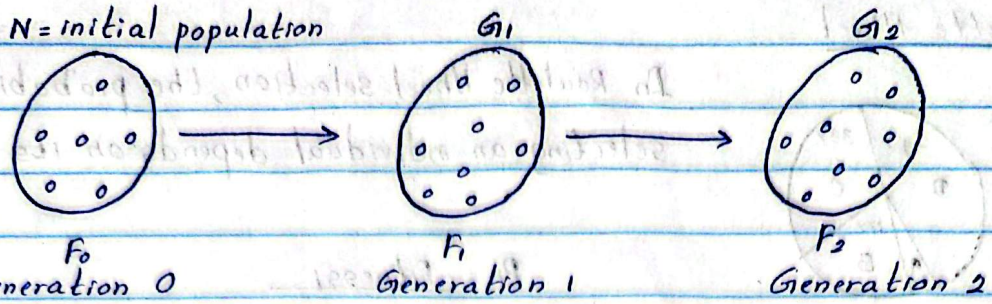


$x = (x_1, x_2, x_3, x_4, x_5, x_6)$ - Chromosome/solution

$z = f(x)$ - Fitness Function

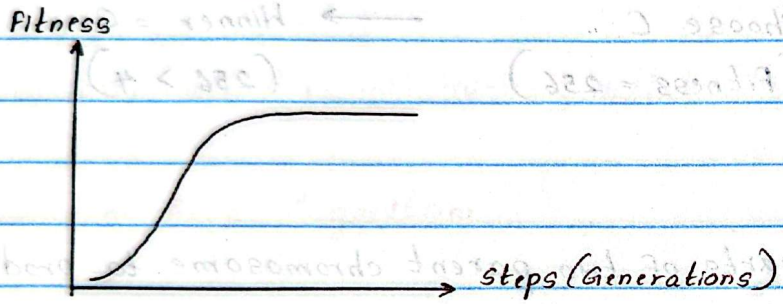
- Each chromosome represents one possible solution.
- The genes (0 or 1) are the decision variables.
- Fitness (F) shows the how good the solution is.

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* Population contains many chromosones (solutions)
* In each generation the population evolves.
* Good solutions survive and produce better solutions.

Fitness over generations



* At the beginning fitness is low.
* As generation increase, better solutions are found and fitness increases.

* After some time, improvement becomes very small and fitness becomes almost constant.
* This means the algorithm has ~~cover~~ converged to a good solution.

Genetic algorithm example

Max $z = x^2$, $0 \leq x < 32$; $x \in z$

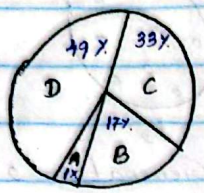
Population size = N = 4

		x	$f(x) = x^2$	
A	0 0 0 1 0	2	4	$4/633 \simeq 1\%$
B	0 0 1 1 1	7	49	$49/633 \simeq 17\%$
C	1 0 0 0 0	16	256	$256/633 \simeq 33\%$
D	1 0 0 1 0	18	324	$324/633 \simeq 49\%$
			633	

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* Selection

Roulette Wheel



In Roulette Wheel selection, the probability of selecting an individual depends on its fitness.

$$P_i = \frac{\text{Fitness}_i}{\text{Total Fitness}}$$

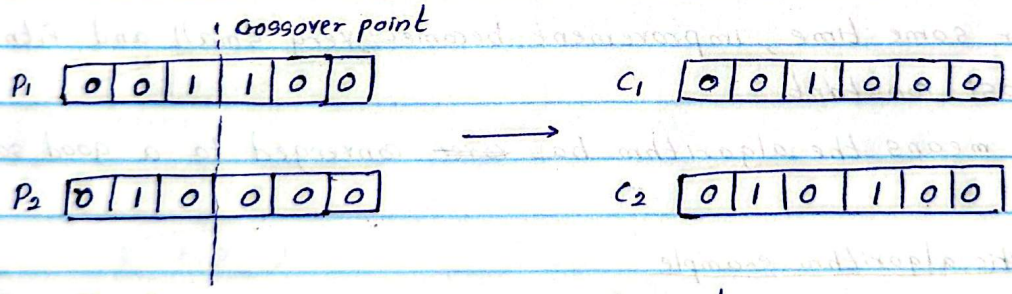
Binary tournament selection

Randomly choose two chromosomes and select the one with better fitness.

Choose A (Fitness = 4) Choose C (Fitness = 256) ⟶ Winner = C (256 > 4)

* Crossover

Crossover combines parts of two parent chromosome to produce children.



Parts after the crossover point are exchanged.

Binary crossover.

* Mutation

Mutation randomly changes one or more genes (0 ↔ 1)

Before mutation (C_1) : 0 0 1 0 0 0

After mutation : 0 0 1 1 0 0

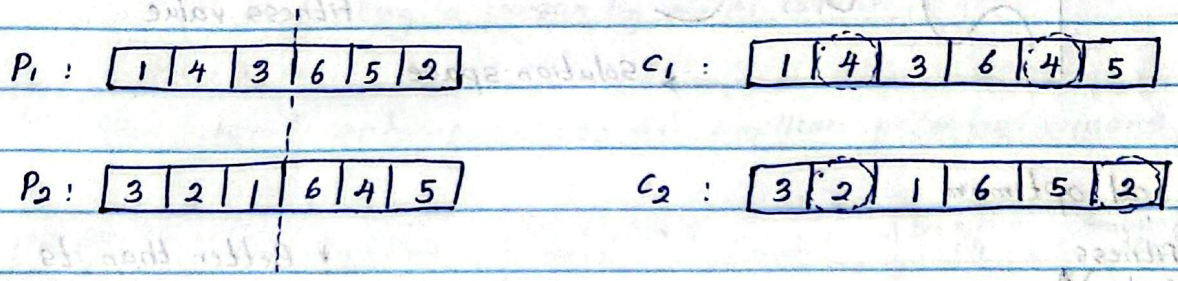
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Purpose :
- Introduce new variation
- Maintain diversity
- Avoid local optimum.

TSP

Travelling salesman Problem - arrange cities in an order to get minimum total distance.



Crossover method is not working well here. So we need another algorithm.

Random Keys Algorithm

This represents a solution using random real numbers, usually between 0 and 1.

P_1 - 0.97 0.68 0.72 0.15 0.48 0.32

P_2 - 0.62 0.38 0.87 0.92 0.16 0.32

Sort the random keys in ascending order to get the city order.

For P_1 :

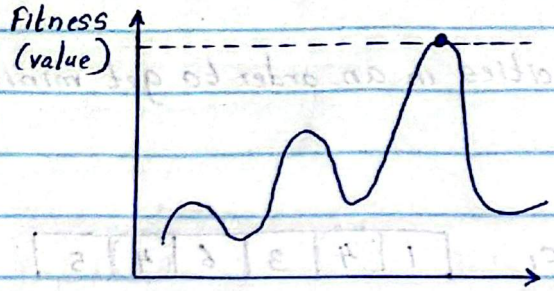
Random key	0.97	0.68	0.72	0.15	0.48	0.32
City No.	1	2	3	4	5	6
Order (Asc)	0.15	0.32	0.48	0.68	0.72	0.97
City order	4	6	5	2	3	1

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Types of optima

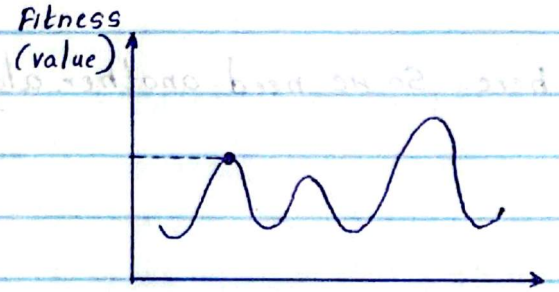
* Optimization is the process of finding the best solution.
* There are different kinds of optima.

1- Global optimum



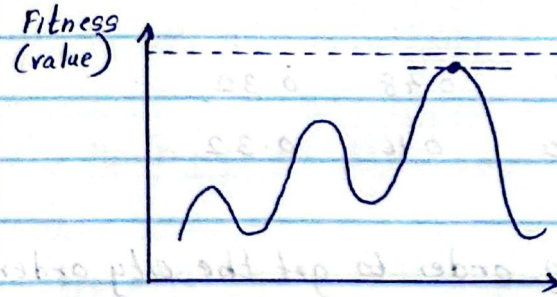
* The best solution among all possible solutions. It has the highest (or lowest) fitness value.

2- Local optimum



* Better than its neighboring solutions but not the best overall solution.

3- Near optimum



* A solution that is very close to the global optimum but not exactly the best.

Stopping criteria examples

- Maximum number of generations reached.
- Desired fitness achieved.
- No improvement for several generations.

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How

Machine Learning & AI Techniques

KDD'98

- KDD'98 is a classic direct marketing dataset used to illustrate how AI/ML techniques are choosen to solve a real business problem.

The business problem

- A charity is deciding how to contact potential donors.
- Cost of contacting a person by phone call - 1₹
- Letter - 25₹
- The dataset contains records of 1 million potential donors.

Donor ID	Characteristics	Donated last year	Amount	Prediction M_1	Amount M_2	Profitability $p \times q$
1		1	1000	1	1200	70% × 1200
2		0	0	1	5000	60% × 500
⋮		1	5000	0	0	0
1M		⋮	⋮	⋮	⋮	⋮

* AI techniques can be used to solve problems by using ML models.

1. Regression
2. Classification
3. Forecasting (Time series prediction)
4. Optimization.

Regression

This is a machine learning technique used to predict a numerical value.

$(x_1, x_2, x_3, \ldots) \longrightarrow$ [Model] $\longrightarrow$?/100

The output is a number, such as the predicted marks for a final examination.

Ex:
- Predicting exam marks
- Predicting donation amount
- Predicting temperature.

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Classification

This is a ML technique used to assign an input to a class/category.

$(x_1, x_2, x_3, \ldots) \longrightarrow$ [ML model] $\longrightarrow$ class

Binary classification

When there are two possible classes, it is called binary classification. (1/0)

* Predict whether someone is going to donate or not.

Multiclass classification

If there are more than two classes, the problem becomes multiclass classification.

class 1, class 2, class 3, class 4

The model predicts one of the four classes.

- Donor prediction example

  The dataset containing approximately 1 million records, with around 250000 identified as donors.

  Model 1 → Classification
  - * Whether someone is going to donate or not.
  - * However classification alone doesn't tell us how much the person will donate.

  Model 2 → Amount of donation
  - * This is a regression problem because the output is a numerical value.

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Prediction	Amount	Probability
1	1200	70% × 1200
1	500	60% × 500
0	0	0

$$\text{Expected value} = \sum_{i=1}^{n} P_i D_i$$

P_i - probability

D_i - predicted donation amount.

70% ⟶ 0.70 × 1200 = 840

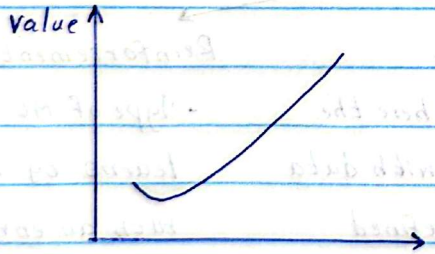
Genetic Algorithm

This is an AI optimization technique inspired by natural selection & evolution.

$$\underbrace{\begin{pmatrix} P_1 & P_2 & P_3 & P_4 & P_5 \\ 1 & 1 & 0 & 1 & 1 \end{pmatrix}}_{P-C} = \overbrace{R}$$

Forecasting

Forecasting is the process of predicting future values using previous observations.



d_1	d_2	d_3	d_4	d_5	...	d_{10}
300	310	312	320	325	...	340

(Extra)

Optimization

finding the best possible solution according to one or more objectives.

minimize objective functions

ex: Minimize cost

Minimize emissions.

Multi objective optimization.

Multi objective optimization is the process of optimizing two or more objective functions at the same time.

CO_2 emission, Fuel cost.

Minimize cost & emission.

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Machine Learning

ML is a branch of AI in which a computer system learns patterns from data and uses those patterns to make predictions, decisions or generate outputs.



ML

Predictive

- Focuses on predicting an outcome from given information.
- Predict future values.
- Predict exam marks

Generative.

- Focuses on generating new content or sequences based on learned patterns.
- Next word prediction in a sequence.

ML

Supervised ML

- Type of ML where the model learns from labelled data.

ex: • Classification
• Regression.

Unsupervised ML

- Type of ML where the model works with data without predefined output labels.

→ Clustering
* Groups similar data items together.

→ Association Rule Mining
* Discovers relationships or associations between items.

Reinforcement Learning

- Type of ML where an agent learns by interacting with an environment & receiving feedback / rewards

States : $s_1, s_2, s_3, \ldots$
Actions : $a_1, a_2, a_3, \ldots$
* The agent learns which action is best for a particular state

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* Reinforcement learning uses a feedback / reward system.
* Feedback can be ⊕ or ⊖
  Positive reward - desirable action
  Negative reward / penalty - undesirable action.

Boston Dynamics - robots. (Robotics company)
* Robots can use reinforcement learning concepts to learn suitable actions through interaction and feedback.

Real coded genetic algorithm.

This represents solutions using real valued numbers rather than only binary values.

ex: (0.25, 1.73, 4.62, 2.15)

This is useful when the decision variables are naturally continuous.

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